Waves

As the wind blows over the ocean, it whips up waves on the surface. The stronger the wind, and the longer it blows, the bigger the waves get. They also grow as they travel, so the biggest waves are the ones that travel long distances over vast oceans, especially the Pacific. Such waves can be destructive when they break on shore. Out on the ocean, they cause less damage, but rare extra-large waves can be dangerous to ships at sea.

WAVE PATTERN

Waves start out as tiny ripples on the water, created by wind blowing over the smooth surface. If the wind keeps blowing, the ripples grow and develop into a confused wave pattern called a chop, with waves of many different sizes and shapes. Gradually, this chaotic effect becomes more ordered, and eventually settles into a regular series of large waves called a swell, which can travel vast distances across the ocean.

▲ RIPPLES
Moving air drags on the surface of the water to push up ripples. These tiny waves are less than 1 in (25 mm) high.

▲ CHOP
Ripples may eventually turn into a chop—a disordered mass of small waves that are up to 20 in (half a meter) high.

▲ SWELL
Over time, waves start rolling across the ocean in a regular swell, with wave crests often towering high above the troughs.

MAKING WAVES
Waves are caused by the way the wind drags on the surface of the ocean. The wind pushes the wave forward, but the water within the wave stays where it is. In fact, each drop of water moves in a circle, rolling forward and then back as the wave passes. This is why objects floating on the water, such as these ducks, stay in the same place as the waves roll under them.